

## 1.6 Vectors

### Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

A vector line can be  $r=a+\lambda b$ . To find intersections, equate components and solve for distinct parameters. A plane  $r\cdot n=d$  has normal vector  $n$ . Dot product tests perpendicularity and angles; cross product gives a perpendicular vector and parallelogram area.

#### Worked example: follow the reasoning

Vectors  $(1,2,0)$  and  $(2,-1,0)$  have dot product  $2-2=0$ , hence they are perpendicular.

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find angle between  $(1,0,0)$  and  $(1,1,0)$ , giving exact value.

[3]

Final answer: \_\_\_\_\_

# Answers and explanation

## Question 1 [3 marks]

Dot product = 1 [1]; lengths 1 and  $\sqrt{2}$  [1];  $\cos \theta = 1/\sqrt{2}$ ,  $\theta = \pi/4$  [1].

### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.